



SHOULD AN ALGORITHM MAKE OUR DECISIONS?



Presented By Group 9

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CONTENT OVERVIEW



Introduction



Case



Problem and Challenges



**Contributing Factors to Problems
in Automated Decision Making**



Analysis and Discussion



Recommendations and Future Direction



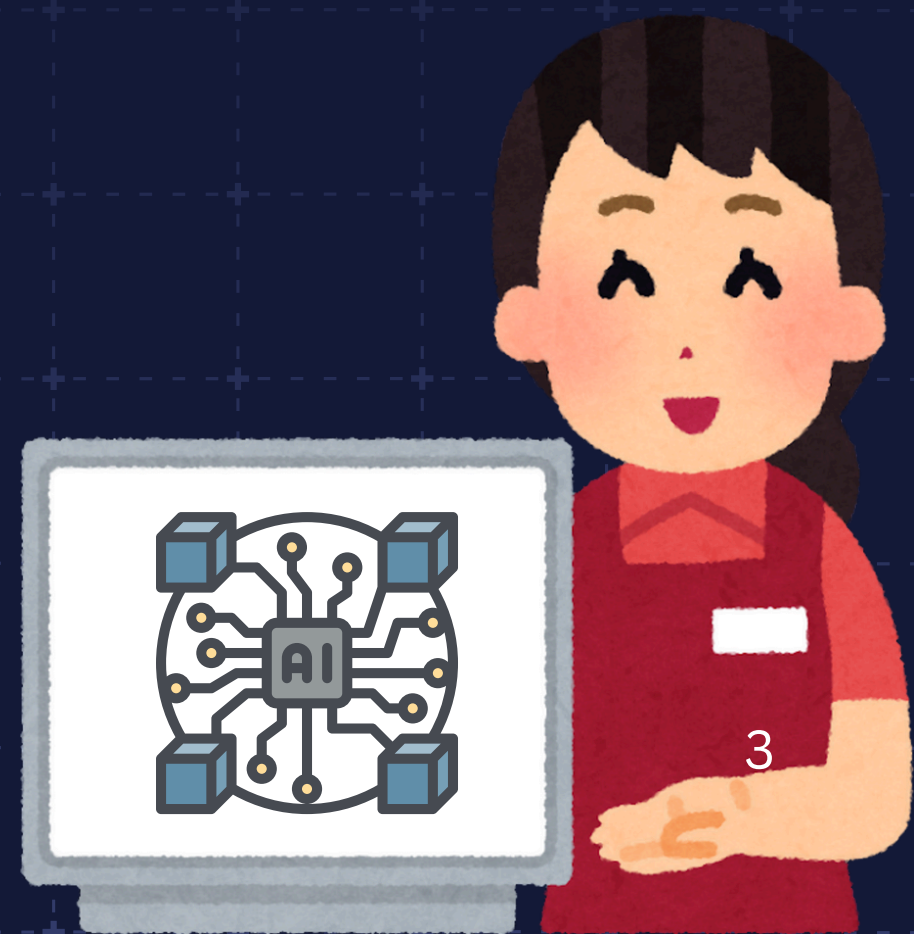
Conclusion



References

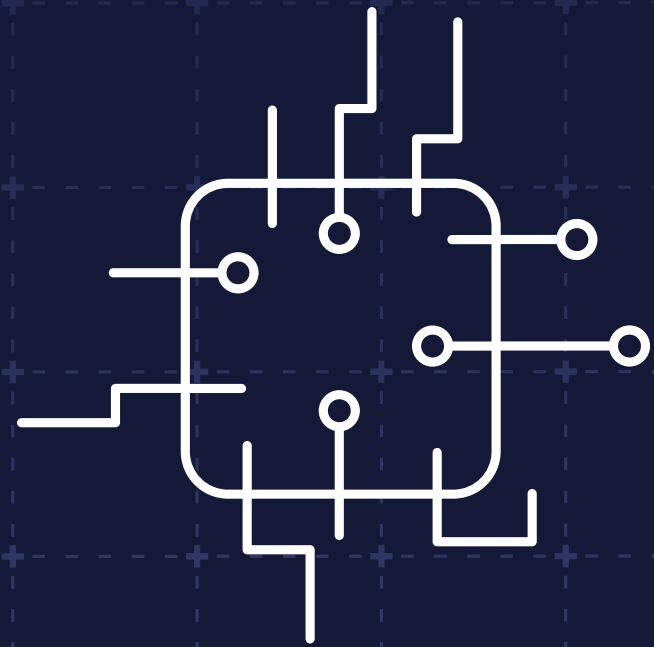


INTRODUCTION



WHAT IS ALGORITHM?

Algorithm



A set of clear and logical steps to solve a problem or complete a task.



Algorithm

GOOD VS BAD

➡ Efficiency and Speed

➡ Problem Solving

➡ Consistency

➡ Personalization

➡ Lack of Transparency

➡ Over-dependence

➡ Privacy Concerns

➡ Potential for Manipulation₅

REAL WORLD CASE

- 1 Darnell Gates, Criminal Risk Assessment (United States)
- 2 Dutch Childcare Benefits Scandal, Fraud Detection Algorithm
- 3 Bristol Youth Risk Scoring, Predictive Risk Assessment (United Kingdom)
- 4 Hospital Patient Prioritization, Healthcare Algorithm (United States)

1.

DARNELL GATES, CRIMINAL RISK ASSESSMENT

- Post-prison risk assessed by algorithm
- Classified as “high risk”, affecting supervision conditions
- No explanation or human engagement provided
- Shows how automated decisions can unfairly shape individual outcomes



Bad Algorithm

2.

DUTCH CHILDCARE BENEFITS SCANDAL, FRAUD DETECTION

- Algorithm flagged applicants using dual nationality & low income
- Thousands falsely accused of fraud
- Families forced to repay benefits, causing social & financial harm
- Demonstrates systemic injustice caused by biased data and weak oversight



Bad Algorithm

3.

BRISTOL YOUTH RISK SCORING, PREDICTIVE RISK ASSESSMENT

- Labeled children as “high risk” using predictive data
- Families not informed or able to challenge
- Lacked transparency, accountability, and consent
- Risk of stigma, discrimination, and long-term harm



Bad Algorithm

4.

HOSPITAL PATIENT PRIORITIZATION, HEALTHCARE ALGORITHM

- Prioritized patients based on predicted healthcare costs
- Systematically underestimated Black patients' health needs
- Used cost as a proxy instead of actual medical need
- Demonstrates how poor design choices can cause bias
- Reinforced existing racial inequalities in healthcare



Bad Algorithm

Problems and Challenges



PROBLEMS



1 Lack of Transparency

2 Algorithmic bias

3 Over-reliance on automated systems

4 limited accountability and appeal



Lack of Transparency



- People are not clearly informed when algorithms affect decisions
- Decision processes are complex or unclear
- Difficult to understand or question outcomes

The Case Examples: Darnell Gates and Bristol Youth Risk Scoring



Algorithmic bias



- Algorithms rely on historical data and proxy indicators
- Indirect factors can reflect existing social inequalities
- Unfair outcomes can occur even without explicit discrimination

The Case Examples: Dutch Childcare Benefits Scandal and Hospital Patient Prioritization



Over-reliance on automated systems



- Algorithmic recommendations are treated as final decisions
- Human judgment and critical evaluation are reduced
- Errors can persist and spread over time

The Case Example: Dutch Childcare Benefits Scandal



Limited accountability and appeal



- Limited ways to review or challenge decisions
- Affected individuals cannot explain their situation
- Power imbalance between institutions and individuals

Contributing Factors to Problems in Automated Decision Making



People Factors

Automation Bias

Over-trusting systems → blindly follow decisions
(e.g. GPS into danger)

Algorithm Aversion

Distrust after one mistake → stop using system even if accurate

Design Gap

Systems reflect creators' background → may fail for others (e.g. facial recognition & darker skin tones)

2 Organizational Factors



Speed over Safety

“Move fast and break things” mindset → AI released without full testing

Accountability Problem

Unclear responsibility for harmful decisions (company, vendor, or employee?) → “problem of many hands”

Data Silos

Departments store data separately → incomplete/dirty data
→ AI produces confident but inaccurate outcomes



Technology Factors



Black Box Problem

Complex models are hard to explain → risky in hiring, healthcare, banking

Feedback Loops

AI reinforces existing patterns → e.g. policing algorithms increasing “recorded” crime in certain areas

Brittleness

Fails in unusual situations → lacks human common sense and adaptability



Legal and Regulatory Factors



Compliance Vacuum

Laws lag behind rapid AI development → risky decisions allowed

Liability Ambiguity

Unclear who is responsible for harm (e.g. self-driving car accidents)

Fragmented Regulations

Different rules across countries, regions, and sectors → hard to ensure compliance

Regulatory Lag

Legislation often reacts after harm occurs → technology evolves faster than laws

5 Environmental Factors

High Energy Consumption

Training large AI models can use as much electricity as hundreds of households per year

Water Usage

Data centers require vast amounts of water for cooling → can strain local resources

Sustainability Concerns

Resource intensity imposes limits on where and how AI can be responsibly deployed

The Role of Algorithm in Decision Making



Algorithms are increasingly used by organizations and public institutions to support decision making.

They can analyze large amounts of data, detect patterns, and provide recommendations quickly. But this raises an important question:

Should algorithms make final decisions on their own
or
Should they serve only as tools to assist human judgment?

Advantages of algorithm

- **Efficient Data Processing**

Algorithms can process large datasets quickly and identify patterns humans might miss.

- **Consistency and Objectivity**

They apply the same criteria across cases, reducing subjective differences.

- **Resource Prioritization**

Helpful in healthcare, probation, and social services to allocate limited resources.

- **Support for Human Judgment**

Helpful in healthcare, probation, and social services to allocate limited resources.

Limitations of Algorithm

- **Predictions, Not Moral Judgments**

Cannot consider ethics, personal circumstances, or social context.

- **Bias and Inequality**

Historical or flawed data can reproduce unfair outcomes for vulnerable groups.

- **Transparency Issues**

Complex “black-box” models make it difficult to understand or challenge decisions.

- **Automation Bias**

People may over-trust algorithmic outputs and reduce critical evaluation.

Algorithms as Decision Support 💡

Algorithms are powerful tools that can enhance efficiency, objectivity, and insight in decision making. However, they cannot replace human judgment, ethical reasoning, or contextual understanding.

To ensure fair and responsible decisions, algorithms should serve as **decision-support** tools while humans remain accountable for interpreting results and considering fairness, ethics, and individual circumstances. Proper oversight and critical evaluation are key to using algorithms safely and effectively.

Recommendation and Future Direction



Human Oversight

Even with powerful algorithms, humans must remain in control of high-stakes decisions that affect people's rights, freedom, or well-being.

Decision makers need to review algorithmic outputs, question unusual results, and consider personal circumstances that algorithms cannot capture.

Clear rules should define when humans are allowed or required to override algorithm recommendations, ensuring that automated scores never automatically decide outcomes.

2 Transparency & Explainability

Transparency helps prevent errors and build trust. Organizations should clearly explain what data is used, how risk scores are generated, and what the results mean.

People affected by algorithmic decisions should know when algorithms influence outcomes and have the right to ask questions or appeal decisions. This accountability ensures mistakes or biases are identified before causing harm.

3 Monitoring and Choosing the Right Metrics

Algorithms need regular audits and monitoring because data or social conditions may change over time, creating unfair outcomes.

Choosing the right performance metrics is crucial, organizations should focus on real goals rather than convenient proxies.

Reviews should involve technical experts, ethicists, policymakers, and representatives of affected communities to ensure fairness, accuracy, and social responsibility.

4 Strong Governance and Responsible Integration

Strong governance, clear regulations, and public engagement are essential for responsible algorithm use.

Governments should set rules for accountability, data use, and human involvement. In the future, algorithms should be thoughtfully integrated with human judgment so decisions remain fair, accountable, and socially responsible.



結論與心得

透過這次的報告和實際案例的分析，我更加認同演算法不適合自己做決定，而應該只是用來支援人類決策的工具。雖然演算法可以提高效率，也能讓結果比較一致，但它沒有道德判斷，也無法真正了解每個人的情況。

如果太依賴演算法，像是荷蘭托兒津貼事件或刑事風險評估的案例，就可能造成不公平，甚至傷害到無辜的人。這次的學習也讓我發現，很多人會因為演算法看起來很科學，就完全相信它的結果，而不再自己思考。

因此，我認為在會影響人自由或未來的重要決定中，人必須負責最後的判斷，而演算法只能用來提供資訊與支援決策，不能取代人類的判斷。



Work Assignment Table

吳美螢	Contributing Factors to Problems in Automated Decision Making, Word Document Processing
陳佳思	Introduction, Real-World Cases
劉晶雯	Problem and Challenges
林珈玟	Analysis and Discussion, Recommendations and Future Direction, Powerpoint Formatting

Peer Evaluation Table

	吳美螢	陳佳思	劉晶雯	林珈紋
吳美螢		10/10	10/10	10/10
陳佳思	10/10		10/10	10/10
劉晶雯	10/10	10/10		10/10
林珈紋	10/10	10/10	10/10	

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**THANK YOU
VERY MUCH!**

